SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I NIKITHAA.B (196526020) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2.0	2.0	2.0	2.0
Explanation	25%	2.0	2.0	2.0	2.0
Application	20%	2.0	2.0	2.0	2.0
Organization	15%	1.5	1.5	1.5	1.5
Accuracy	10%	1.0	1.0	1.0	1.0
Clarity	5%	0.5	0.5	0.5	0.5

Total Marks 36 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

COT - I

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A company transfers a 150MB design file over a network with a bandwidth of 50 Mbps. The transport layer divides the file into 1500 bytes segments, and the Data link layer divides the adds 40 bytes of header and trailer to each frame. Calculate the total number of segments and frames, total number of segments and frames, total overhead, and transmission time. Explain how both layers ensure reliable communication

36/40

Sol

Given,

File Size = 150MB
Network bandwidth = 50 Mbps
Segment Size = 1500 bytes
Data link layer Overhead = 40 bytes per frame
Propagation delay = Ignored

1MB = 1,000,000 bytes
1Mbps = 1,000,000 bits per second.

150MB = 150,000,000 bytes

Number of Segments (n) = Total file size / Segment size

$$n = \frac{150,000,000}{1500}$$

n = 100,000 Segments

Each segment is encapsulated into one data link layer frame

Therefore,

Total frames = 100,000

Each frame contain 40 bytes of additional header and trailer information

Total overhead (N) = Number of frames x Overhead per frame

(N) = 100,000 x 40

$N = 4{,}000{,}000$ bytes

$N = 4MB$

∴ Total Overhead (N) = 4MB

Total data transmitted:

$= 150{,}000{,}000 + 4000{,}000$

$= 154{,}000{,}000$ bytes

Bytes into bits

$= 154000{,}000 \times 8 =$

$= 1{,}232{,}000{,}000$ bits

Transmission time:-

$$T = \frac{\text{Total bits}}{\text{Bandwidth}}$$

$$T = \frac{1{,}232{,}000{,}000}{50{,}000{,}000}$$

$T = 24.64$ Seconds.

Role of transport and Data link layers in Reliable Communication

Transport layer:-

The transport layer divides application data into smaller segments and reassembles them at the destination. Protocols such as TCP provide reliability through sequence numbers, acknowledgements, error detection and retransmission of lost segments.

Data link layer:-

The Data link layer divides application encapsulates segments into frames for transmission over a local link. It uses frame addressing and error detection, commonly through a frame check sequence, to identify corrupted frames. Depending on the link protocol, damaged frames may be retransmitted.

A transport layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows and the total retransmissions when one frame in every window is lost. Explain how flow control improves communication efficiency.

Given,

Sliding window size = 6 frames
user data per frame = 1024 bytes
Total frames transmitted = 48
lost frames per window = 1

Number of window (N) = $\frac{\text{Total frames}}{\text{Window size}}$

$N = \frac{48}{6}$

Number of window (N) = 8 windows

Total retransmission (R) = Number of windows x Lost frames per window

$= 8 \times 1$

$R = 8$ retransmissions

Total user data = Total frames x Data Per frame

$= 48 \times 1024$

$= 49,152$ bytes

Therefore,

Total user data = 48KiB

The sliding window mechanism improves Communication efficiency in the followings ways:

Multiple - frame transmission:

It allows several frames to be transmitted before acknowledgements are received, reducing unnecessary waiting

Prevents receiver Overload:

It limits the amount of unacknowledged data according to the receiver's Capacity.

Improves bandwidth utilization:

It helps keep the network busy instead of waiting after every individual frame.

Support recovery:

Lost frames can be retransmitted according to the product recovery mechanism.

Reduces unnecessary delays:

Efficient, transmission and acknocoledgement handling improve Overall Communication performance.

The Sender requires 8 transmission windows and under the question assumption of one lost frame per window, 8 transmissions. flow control helps maintain efficient data transmission while preventing the receiver from being Overwhelmed.

A 12000 byte IP Packet is transmitted through a router with an MTU of 1500 bytes. Each fragment requires of a 20-byte IP header. Calculate the number of fragments and total header Overhead. Explain how the Network layer ensures Successful delivery of fragmented Packets.

IP Packet Size = 12000 bytes

Maximum Transmission Unit (MTU) = 1500 bytes

IP header per fragment = 20 bytes

Maximum data per fragment (F) = MTU − IP header

(F) = 1500 − 20

(F) = 1480 bytes

Original payload Size:

= 12000 − 20

= 11,980 bytes

Number of fragments = $\frac{11,980}{1480}$

= 8.0946

Number of fragments ≈ 9

first 8 fragments : 1480 bytes of payload each

final fragment = 11,980 − (8 × 1480)

= 140 bytes of payload.

Total header Overhead = Number of fragments × Header Size

= 9 × 20

= 180 bytes

Total transmitted IP data = Original Payload + fragmentation header

= 11,980 + 180

= 12,160 bytes

The Network layer uses the following mechanisms to handle fragmented Packets:-

1) Fragmentation:-

It divides an IP packet into smaller fragments when the packet exceeds the MTU, where fragmentation is permitted.

2) Identification:-

IPV4 fragments from the same original packet carry a common identification value.

3) Fragment Offset:

This field indicates the position of each fragments' payload in the Original packet.

4) More fragments flags:-

It indicates the position whether additional fragments follow.

5) Reassembly:

The destination uses the identification, offset and fragment flags to reassemble the Original packet.

6) Error handling:-

If fragments are missing or cannot be reassembled before the reassembly timeout the original packet cannot be completed. Higher-layer protocols such as TCP may arrange retransmission.

A multimedia application transfers 600 MB Video file. The Session layer inserts a Synchronization Checkpoint after every 60MB of transmitted data. A failure occurs after 390 MB has been transffered. Determine the number of checkpoints created before failure and the amount of data that must be retransmitted after recovery. Explain the importance of Synchronization in the Session layer.

Total Video file Size = 600MB

Checkpoint interval = 60MB

Data transfered before failure = 390MB

Number of checkpoint = $\frac{\text{Data transfered}}{\text{checkpoint interval}}$

$= \frac{390}{60}$

Number of Checkpoints	= 6.5

Since checkpoint are created at every completed 60 MB interval, the checkpoints occurs at 60MB, 120MB, 180MB, 240MB, 300 MB, 360MB

Therefore,

6 checkpoints.

The 390 MB transfer point is between the checkpoints at 360 MB and 420 MB

Data to retransmit = Data transfered before failure - Last checkpoint

$= 390 - 360$

$= 30MB$

Importance of Synchronization in the Session layer

The Session layer of the OSI model establishes, manages and terminates communication Session between applications. Synchronization checkpoints help applications recover from interruption without restarting the entire transfer.

It main functions and benefits include:

* checkpoint creation : It establishes recovery points at regular intervals during a long data transfer
* Error recovery : After a failure, the application can resume from the most recent Vaild checkpoint instead of Starting again.
* Reduced retransmission : only the data transferred after the last checkpoint need to be retransmitted reducing unnecessary network traffic.
* improved efficiency : checkpoints Save time and bandwidth during large file transfers.
* Session Continuity : Synchronization helps maintain the progess of a Communication session after a temporary interruption